

ESP32 Gate Controller

A production-grade dual-leaf gate controller built on ESP32 with FreeRTOS, featuring a web dashboard, local display menu, HomeKit integration (planned), and full NVS-backed configuration.

Features

- **Dual-leaf gate control** — independent relay sequencing with configurable delay between leaves
- **4 reed switch inputs** — dedicated open/close end-of-travel sensors per leaf (4 total)
- **FreeRTOS multi-task architecture** — gate control, input handling, web server, display, and monitor running as independent tasks across both ESP32 cores
- **Safety layer** — hardware interlocks prevent simultaneous activation of opposing relays; configurable safety timeout force-stops on stall
- **State machine** — 6 states: CLOSED, OPENING, OPEN, CLOSING, STOPPED, ERROR with full open/stop/reverse cycle on single button press
- **Sequence inversion** — configurable leaf order (L1→L2 or L2→L1) without code changes
- **Web dashboard** — tabbed interface (Status / Config / Log) served directly from ESP32; accessible from any browser on the local network
- **WiFi captive portal** — AP mode configuration page when no credentials are stored
- **NVS configuration** — all parameters persisted across reboots (leaf delay, safety timeout, alert threshold, sequence order, language)
- **Event log** — FIFO of 50 events stored in NVS with ISO timestamp (yyyymmddhhmmss); populated from NTP when available
- **Optional ST7735 display** — 160x128 TFT with animated state display, reed status, clock, IP address; auto-detected at boot
- **3-button menu** — full configuration menu on display without WiFi or mobile device

- **Bilingual UI** — English / Portuguese switchable at runtime
- **NTP time sync** — UTC-3 (Campinas, BR); timestamps fall back to placeholder when offline

Hardware

Bill of Materials

Component	Qty	Notes
ESP32 Dev Module (38-pin)	1	ESP32-D0WD-V3 or equivalent
4-channel relay module (optocoupler)	1	5V coil, active-HIGH
Reed switch (magnetic)	4	2 per leaf: open + close end-of-travel
ST7735 TFT 1.8" 128x160 SPI	1	Optional — firmware detects presence
Tactile push button	3	+, -, OK (menu navigation)
10kΩ resistor	4	Pull-up for reed switches (GPIO 32-35)
Hi-Link HLK-5M05	1	220VAC → 5VDC isolated PSU (production)
MOV varistor	1	Surge protection on 220VAC input

Pin Map

Signal	GPIO	Notes
Relay L1 Open	25	Motor 1 open direction
Relay L1 Close	26	Motor 1 close direction
Relay L2 Open	27	Motor 2 open direction
Relay L2 Close	14	Motor 2 close direction
Reed Open L1	34	LOW = leaf 1 fully open

Reed Close L1	35	LOW = leaf 1 fully closed
Reed Open L2	32	LOW = leaf 2 fully open
Reed Close L2	33	LOW = leaf 2 fully closed
Button GATE	22	Toggle gate (short press)
Button MINUS [-]	16	Menu navigation / decrement
Button PLUS [+]	17	Menu navigation / increment
Button OK	0	Menu select / WiFi reset (3s hold at boot)
Display CS	5	ST7735 SPI chip select
Display DC	2	ST7735 data/command
Display RST	4	ST7735 reset
Display MOSI	23	SPI MOSI
Display SCK	18	SPI clock

Architecture

FreeRTOS Task Layout

Core 0

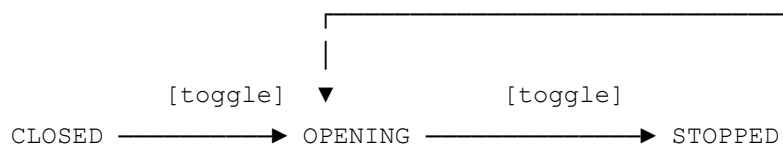
```
taskMonitor (prio 1)
taskWeb      (prio 1)
```

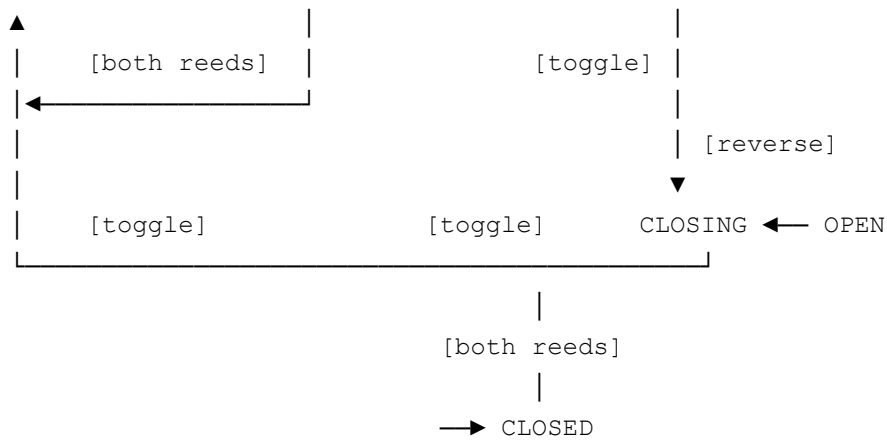
Core 1

```
taskGateController (prio 2)
taskInputHandler   (prio 2)
taskDisplay         (prio 1)
```

WiFi stack and LwIP TCP/IP run natively on Core 0. Gate control and display are isolated on Core 1 to prevent network latency from affecting relay timing or UI responsiveness.

Gate State Machine





Timeout on OPENING or CLOSING → ERROR state. Any toggle in ERROR resets to CLOSED.

Relay Sequencing

Opening (normal order):

```

t=0          Relay L1_OPEN ON
t=0+         Monitor REED_OPEN_L1 → LOW: Relay L1_OPEN OFF
t=leafDelay  Relay L2_OPEN ON  (independent of L1 reed)
t=leafDelay+ Monitor REED_OPEN_L2 → LOW: Relay L2_OPEN OFF
t=T          Safety timeout: all relays OFF → ERROR
  
```

Closing reverses leaf order (L2 first, then L1 after delay).

Hardware safety: opposing relays of the same motor (e.g. L1_OPEN and L1_CLOSE) are never activated simultaneously. A 10ms dead-time is enforced between deactivating one and activating the other.

Software

Dependencies

Library	Version	Purpose
arduino-esp32	≥ 2.0	ESP32 Arduino framework
TFT_eSPI	Bodmer	ST7735 display driver
OneWire	Jim Studt	(temperature project, separate)

DallasTemperature	Miles Burton	(temperature project, separate)
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Configuration (NVS)

All parameters are editable via the web dashboard or display menu and persist across reboots:

Parameter	Default	Description
leafDelay	3000 ms	Delay before activating second leaf
safetyMs	20000 ms	Max movement time before force-stop
alertMin	10 min	Alert if gate stays open longer
seqInv	false	Invert leaf sequence (L2 leads)
lang	EN	UI language (EN / PT)

Event Log

50-entry FIFO stored in NVS. Events: OPENED, CLOSED, STOPPED, TIMEOUT, OPEN>LIMIT, ERROR. Timestamps in `yyyymmddhhmmss` format populated from NTP; `0000000000000000` when offline.

Web Interface

Served at the device IP on port 80. Three tabs:

- **Status** — live gate state, reed indicators, toggle button, WiFi reset
- **Config** — all parameters with live save
- **Log** — event history with clear option

When no WiFi credentials are stored, the device starts in AP mode (`ESP32 Gate`) and serves a captive portal for credential entry.

Display Menu

Activated by pressing OK. Navigated with +/- and OK. Returns to idle after 30s of inactivity.

Menu

```
|— Gate Control
|   |— Toggle Gate
|— Configuration
|   |— Leaf Delay
|   |— Safety Timeout
|   |— Alert Open
|   |— Sequence
|   |— Language
|— Event Log
|   |— [scrollable entries]
|   |— Clear Log
|   |— Back
|— WiFi
|   |— Reset Credentials
|   |— Back
|— Exit
```

Production Notes

 This controller switches 220VAC motor circuits. The following are mandatory for any installation:

- Galvanic isolation between ESP32 (3.3V logic) and relay contacts (220VAC)
- Use relay modules with built-in optocouplers (e.g. EL817)
- Isolated switching PSU (e.g. Hi-Link HLK-5M05) — never use a non-isolated supply
- MOV varistor on 220VAC input for surge protection
- Physical barrier between HV and LV sections inside enclosure
- Fuse on 220VAC input

Roadmap

- MQTT integration for remote access outside local network
- HomeKit / Siri / CarPlay widget via HomeSpan
- OTA firmware update

- PCB design (KiCad)
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Embedded systems, IoT, firmware development

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